

Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 17: Forest: Our Lifeline

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	17 — Forest: Our Lifeline
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Traps: confusing a tree's crown with the whole-forest canopy; thinking decomposers attack living trees or merely "destroy" matter rather than recycle it into humus; assuming removing a top predator helps the forest; reading a food web as a single straight chain; believing energy or nutrients flow upward to the Sun; expecting a cleared forest to grow back overnight; crediting forests with "making" oxygen at night; mistaking the layer with most life for the layer with most light. Do not write on this paper — mark answers on your answer sheet.

1. A scientist studies a patch of land for ten years and decides it qualifies as a forest. Which observation would be the **STRONGEST** single piece of evidence for that decision?

- (A) A very large count of tall trees of one single species packed close together
- (B) The presence of at least one large predator such as a tiger somewhere in the area
- (C) Many different species of plants, animals and micro-organisms living in interlinked layers, persisting and renewing themselves over the whole period
- (D) A thick covering of green grass spread evenly across the open ground

2. Two students describe parts of a forest. Riya points to the leafy roof formed where the branches of **MANY** tall trees meet and overlap. Sam points to the branchy top of **ONE** single tree. What are they naming, in that order?

- (A) Riya names the crown; Sam names the canopy
- (B) Both are naming the canopy, just from different sides
- (C) Riya names the understorey; Sam names the canopy
- (D) Riya names the canopy; Sam names the crown

3. A short, shade-tolerant tree grows in the dim space between the forest floor and the high treetops. Which layer does it belong to, and why does it tolerate low light?

- (A) The canopy — it must reach the very top to get any light at all
- (B) The forest floor — only the floor has shade plants and there is no layer above it
- (C) The crown — every short tree forms its own crown and needs no light
- (D) The understorey — it is adapted to live on the dim light that passes through the canopy above

4. On the floor of a healthy forest, fallen leaves never pile up into a permanent mountain even though leaves fall every year. The best explanation is that:

- (A) Decomposers steadily break the fallen leaves down into humus at about the rate new leaves arrive
- (B) Wind blows every fallen leaf out of the forest before it can settle
- (C) The canopy trees suck their old leaves back up to use them again
- (D) Falling leaves dissolve instantly the moment they touch the ground

5. Humus formed on the forest floor is valuable mainly because it:

- (A) Adds extra oxygen straight into the air for animals to breathe
- (B) Acts as a hard, waterproof seal that stops any rain from entering the soil
- (C) Returns nutrients to the soil so that plant roots can take them up and grow new plants
- (D) Provides ready-made food that animals eat directly in place of plants

6. In the chain grass → grasshopper → frog → snake, the grasshopper plays which role, and on what does it depend for energy?

- (A) A herbivore that depends ultimately on the grass, a producer, for its energy
- (B) A producer that makes its own food and depends on the Sun directly
- (C) A carnivore that depends on the frog above it for energy
- (D) A decomposer that gets energy from breaking down the dead snake

7. A forester wants to show why the number of tigers in a forest is far smaller than the number of deer, and the deer far fewer than the grass plants. The correct reasoning is:

- (A) Tigers simply choose to live alone while deer prefer to gather in large herds
- (B) Only a small part of the energy is passed on at each step up the chain, so fewer organisms can be supported higher up
- (C) Grass reproduces but deer and tigers cannot have young, so their numbers stay low
- (D) Tigers need almost no food, so very few of them can survive on what is available

8. Why is a food WEB a more realistic picture of a forest than a single food chain?

- (A) A web has no producers, which is closer to how a real forest works
- (B) Most animals eat more than one kind of food and are eaten by more than one kind of animal, so many chains cross and link

- (C) A web always contains exactly the same number of organisms in every forest
- (D) A web shows that energy can flow backwards from carnivores to plants

9. Read this forest web. Leaves are eaten by caterpillars and by beetles. Caterpillars are eaten by warblers (small birds). Beetles are eaten by shrews. Warblers and shrews are both eaten by owls. Which statement is true of the OWL in this web?

- (A) It draws food from two different chains, since warblers and shrews start from different plant-eaters
- (B) It feeds on only one chain, because warblers and shrews are really the same animal
- (C) It is a producer, because it sits at the very top of the web
- (D) It eats leaves directly, since every top animal also feeds on plants

10. In the same web (leaves → caterpillars → warblers; leaves → beetles → shrews; warblers & shrews → owls), a fungus disease wipes out almost all the CATERPILLARS. The most likely FIRST effect is:

- (A) Owls die out at once, because the caterpillar was their only possible food
- (B) Warblers lose their main food and tend to decline, while leaves eaten by caterpillars are spared
- (C) The leaves disappear immediately, because caterpillars were protecting them
- (D) Beetles vanish straight away, since they depended on the caterpillars

11. Continuing the same web: if the OWLS are all removed, the most likely chain of effects is:

- (A) The leaves vanish at once because owls were the forest's producers
- (B) Nothing changes anywhere, since the owl was at the top and ate nothing important
- (C) Warblers and shrews face fewer predators and may increase, which can then press down caterpillar and beetle numbers
- (D) Caterpillars and beetles rise immediately because owls used to eat them

12. A child says, "If we just remove all the decomposers, the forest will stay tidy and the trees will grow better." The best correction is:

- (A) The child is right — decomposers only cause rot and steal food from the trees
- (B) It makes no difference, because trees feed only on sunlight and need no soil nutrients
- (C) Without decomposers, dead matter would pile up and its nutrients would stay locked away, so the soil would grow poorer and trees would grow worse
- (D) Trees would grow better, since decomposers attack and kill living tree roots

13. The nutrient cycle of a forest runs: soil → plants → animals → dead bodies and dung → decomposers → soil. Which single break in this loop would most directly stop nutrients reaching the soil again?

- (A) Adding more sunlight, which would speed the whole cycle up too much
- (B) Removing the herbivores, which are the only link that creates humus

- (C) Adding rain, which carries every nutrient straight out to the sea
- (D) Removing the decomposers, so dead bodies and dung are never broken back down into humus

14. Why does cutting the CANOPY trees of a forest harm the herbs and small floor animals far more than just losing those individual big trees?

- (A) The canopy was the only source of soil nutrients for the lower plants
- (B) The canopy shelters the layers below, so removing it suddenly exposes shade plants and floor animals to harsh sun, heat and beating rain
- (C) Floor animals breathed only the air trapped inside the canopy leaves
- (D) Removing the canopy lowers the floor temperature far below freezing

15. In a forest, the GREEN PLANTS are the producers. The single best reason they hold this role is that they:

- (A) Capture sunlight and use it with water and carbon dioxide to make their own food, the energy source for everything else
- (B) Are simply the most numerous living things in the forest by count
- (C) Provide the wood and gum that humans collect from the forest
- (D) Take in oxygen and dead matter from the soil to build their bodies

16. A forest plot has these animals: deer (eats shrubs), rabbit (eats herbs), fox (eats rabbits), wild dog (eats deer and foxes). Which animal is the only one here that gets food from TWO other animals?

- (A) The fox, which eats both rabbits and deer
- (B) The deer, which eats both shrubs and rabbits
- (C) The rabbit, which eats both herbs and foxes
- (D) The wild dog, which eats both deer and foxes

17. Using the same plot (deer→shrubs, rabbit→herbs, fox→rabbits, wild dog→deer & foxes): if a disease kills almost all the RABBITS, the most directly affected animal is:

- (A) The fox, because rabbits are its food
- (B) The deer, because deer and rabbits eat the same plants
- (C) The wild dog, because the wild dog hunts rabbits directly
- (D) The shrubs, because rabbits used to protect the shrubs

18. In the same plot, the dead bodies of the wild dog, the fox and the fallen shrubs are eventually returned to the soil by:

- (A) Decomposers such as fungi and bacteria on the forest floor
- (B) The deer, which graze on dead animal bodies
- (C) The Sun, which dries and recycles them into the soil
- (D) The remaining rabbits, which carry nutrients down to the roots

19. Forests are called the "green lungs" of the Earth. Which description of what their plants do BY DAY best fits that name?

- (A) They take in oxygen and breathe out carbon dioxide, exactly like animal lungs
- (B) They store large amounts of water as a mist held in the air
- (C) They give out neither oxygen nor carbon dioxide and simply filter dust
- (D) They take in carbon dioxide and release oxygen, helping keep the air's gas balance

20. A patch of dense forest on a hill is completely cleared. In the next monsoon, the bare slope shows much more soil being washed away than before. The best reason is that, with the trees and roots gone:

- (A) The Sun now shines on the slope and bakes the soil so hard it floats away
- (B) Rainwater has become heavier and so it scours away more soil than before
- (C) There is nothing left to grip the soil and slow the rain, so water rushes down carrying soil with it
- (D) The missing trees used to drink the soil, and now the soil has nowhere to go

21. Why does a forested catchment release water to a stream slowly and steadily, while a deforested one gives sudden floods then runs dry?

- (A) The forest's trees pump water up into the clouds and time its release to the stream
- (B) Forest streams are simply built deeper, so they never flood or dry up
- (C) The bare land soaks up far more water than forest soil, so it floods on purpose
- (D) The forest's roots and leaf litter let rain soak into the ground and feed the stream gradually; bare land lets rain rush off all at once and store nothing

22. A village finds that after the nearby forest was cut, summers feel hotter and the rains are less reliable. Through which everyday process did the forest most help with local moisture and rainfall before it was lost?

- (A) Trees released water vapour through their leaves (transpiration) and shade, adding moisture to the air and helping bring rain
- (B) Trees pulled rain clouds down from the sky with their tall trunks
- (C) Trees burned in the sun and the smoke seeded the clouds with rain
- (D) Trees absorbed all the heat from the Sun so it could never warm the land

23. Cutting down a large area of forest tends to RAISE the level of which gas in the air, and why?

- (A) Oxygen, because with no trees nothing is left to use up the oxygen
- (B) Carbon dioxide, because fewer living trees are left to take it in during photosynthesis
- (C) Nitrogen, because removing trees releases nitrogen stored in their wood
- (D) Water vapour only, because cut wood gives off steam as it dries

24. Where is most of the carbon that a growing forest takes from the air actually stored?

- (A) In the empty air between the trees, packed tightly together
- (B) In the wood, leaves and soil, where it is built into plant tissue and humus
- (C) In the fast-flowing streams that run through the forest
- (D) In the fur and feathers of the forest animals

25. Which list contains ONLY things that are genuine products obtained from forests?

- (A) Honey, gum, medicinal herbs and timber
- (B) Plastic, petrol and timber
- (C) Glass, gum and honey
- (D) Timber, medicines and cement

26. A boy claims, "A cleared forest is no problem — we can simply grow a brand-new full forest in a few months whenever we want." Why is this wrong?

- (A) It is correct; planting seeds gives a complete grown forest within one season
- (B) A cleared forest never grows back at all, so the time does not matter
- (C) A forest builds up slowly over many years as herbs, then shrubs, then trees re-establish and the soil and community recover — it cannot appear in a few months
- (D) Forests grow back overnight, but only if a tiger is released into the area first

27. A keystone predator like the tiger keeps deer numbers in check. Predict what overgrazing by too many deer would do to the forest if the tigers disappeared:

- (A) More deer means more dung, which instantly turns the whole area into rich new forest
- (B) Nothing changes, because plants and animals in a forest do not affect each other
- (C) The extra deer would start eating tigers and balance their own numbers
- (D) The deer multiply and strip the young plants and shrubs, so the forest struggles to renew itself

28. Which single statement correctly captures why removing just ONE species can ripple through a whole forest?

- (A) Because every species in a forest lives completely on its own and never affects another
- (B) Because the species are interdependent — they feed on, shelter or control one another, so a change in one passes along the links to many others
- (C) Because all the species are exactly the same and so behave identically
- (D) Because only the largest animal matters and the rest have no connections at all

29. A field of wheat has millions of plants of a single kind grown for one season, then harvested. Give the best reason it is NOT a forest:

- (A) A field is simply smaller than a forest, and size is the only difference
- (B) A field has no animals or insects in it at all, while a forest does
- (C) A forest is a many-species, many-layered community that persists and renews over many years, not a single crop grown and cleared in one season
- (D) A field cannot be called a forest unless it contains at least one large tree

30. In the chain leaves → caterpillar → small bird → snake, what is the snake's role, and where did the energy in its body originally come from?

- (A) A herbivore, whose energy came directly from eating the leaves
- (B) A producer, which made the energy itself from sunlight
- (C) A carnivore at the top of this chain, whose energy traces all the way back to the leaves
- (D) A decomposer, which got its energy by breaking down dead small birds

31. Which pairing of a forest ROLE with the RIGHT organism is correct?

- (A) Producer — tiger
- (B) Carnivore — grass
- (C) Herbivore — fungus
- (D) Decomposer — mushroom (a fungus)

32. A forest captures carbon, holds soil, recharges groundwater, refreshes the air and shelters wildlife — all at once. The deepest reason a forest can deliver so many services together is that it is:

- (A) Simply a very large pile of timber waiting to be cut and sold
- (B) A collection of unrelated trees that happen to grow near one another
- (C) A man-made arrangement that has to be rebuilt by people every year
- (D) A living, interconnected system whose plants, animals, soil and micro-organisms work together, each part supporting the others

33. Why is forest soil usually far richer and more fertile than the soil of a bare, cleared hillside?

- (A) Decomposers in the forest constantly turn fallen leaves and dead matter into humus, while litter and roots also hold the rich soil in place
- (B) The canopy blocks all rain, so the soil is kept dry and that makes it fertile
- (C) Animals plough the forest soil with their hooves and add no other material
- (D) There is no sunlight on the forest floor, and darkness by itself makes soil rich

34. Spell out the chain of events most likely to follow large-scale deforestation of a region:

- (A) Fewer trees → more oxygen in the air and richer, deeper soil over time
- (B) Fewer trees → steadier streams, more rainfall, and cooler, wetter summers
- (C) Fewer trees → more soil erosion, less reliable rainfall, lost wildlife homes, and more carbon dioxide left in the air
- (D) Fewer trees → groundwater rises everywhere and floods stop completely

35. A monkey leaps among the high, sunlit branches and a fruit-eating bird nests up there too. In which forest layer do these animals mainly live, and why is it suited to them?

- (A) The canopy, because its high branches catch the most sunlight and bear the leaves and fruit they feed on
- (B) The forest floor, because that is where the most light and food are found
- (C) The understory, because the tallest animals always stay in the middle layer
- (D) Inside the soil, because canopy animals burrow to escape the sunlight

36. Earthworms, beetles and fungi are found mainly on or just under the forest floor rather than up in the canopy because the floor:

- (A) Receives the most direct sunlight, which these animals need to make food
- (B) Is the only layer that has any oxygen for small animals to breathe
- (C) Is colder than ice, which is the condition fungi and worms prefer
- (D) Is where dead leaves and dead matter collect, giving these decomposers and ground feeders their food and damp shelter

37. Sunlight that filters down through gaps in the canopy matters to the lower layers because it:

- (A) Is harmful to all lower plants and should be blocked out completely
- (B) Only warms the floor animals and plays no part in feeding the plants
- (C) Gives shade-loving understorey and floor plants just enough light to make their own food and survive
- (D) Removes the need for the lower plants to take up any water or nutrients

38. Of the canopy, the understorey and the forest floor, which receives the LEAST sunlight, and which therefore tends to be home to the most shade-loving and decomposing organisms?

- (A) The canopy receives the least light because it is shaded by the clouds
- (B) All three layers receive exactly equal sunlight, so none is shadier
- (C) The understorey receives the least light because it is buried in the soil
- (D) The forest floor receives the least light and hosts the most shade-loving plants and decomposers

39. Which organism in a forest is a true DECOMPOSER, and what exactly is its job?

- (A) A mould fungus, which breaks down dead leaves and dead animals into humus
- (B) A deer, which decomposes the grass it grazes on inside its stomach
- (C) A mango tree, which decomposes sunlight into food for the forest
- (D) A hawk, which decomposes the small birds it catches and eats

40. Web A: grass → grasshopper → frog → snake → hawk. Web B: grass → rabbit → fox. A naturalist notes that both webs share the same first link. What does this tell us about where ALL the energy in both webs begins?

- (A) It begins with the hawk and the fox, the top animals, and flows downward to the grass
- (B) It begins separately in each animal, which makes its own energy as it needs it
- (C) It begins with the grass, a producer that captured the Sun's energy, on which every animal in both webs ultimately depends
- (D) It begins in the soil's humus, which the animals eat directly to get their energy

41. In Web A above (grass → grasshopper → frog → snake → hawk), the grasshopper is a herbivore and the frog, snake and hawk are carnivores. Which animal in this chain

depends on the SMALLEST share of the Sun's original energy?

- (A) The grasshopper, because it is the smallest-bodied animal in the chain
- (B) The grass, because producers always get the least energy of all
- (C) The hawk, because energy is lost at each step and it sits at the top of the chain
- (D) The frog, because animals in the middle always receive the least energy

42. A child wonders why a forest does not run out of soil nutrients even though plants keep taking nutrients up year after year. The best answer is:

- (A) Decomposers keep returning nutrients to the soil from dead leaves, dung and dead animals, replacing what the plants take
- (B) Plants actually add brand-new nutrients to the soil from the sunlight they absorb
- (C) The nutrients never really leave the soil because plants only borrow them briefly
- (D) Rain washes in a fresh supply of nutrients from the sky every single day

43. A forest pond web: algae → small fish → kingfisher (a bird). On land beside it: leaves → deer → tiger. A single decomposer community on the floor breaks down everything that dies in BOTH. What does this shared decomposer link show about a forest?

- (A) Water chains and land chains are completely separate and never connect in a forest
- (B) Different food chains are tied together through shared decomposers that recycle the dead from all of them back into one soil
- (C) Decomposers only work on land animals and ignore anything that dies in water
- (D) Each chain has its own private decomposers that never touch another chain's dead

44. If the small fish in that pond web (algae → small fish → kingfisher) almost all die from pollution, the most directly affected animal and plant are:

- (A) The deer and tiger on land collapse at once, since they shared the fish
- (B) The kingfisher loses its food and tends to decline, while the algae the fish ate are no longer grazed and may build up
- (C) The algae die immediately, because the fish were their only food source
- (D) The kingfisher increases, because fewer fish means cleaner water for it

45. Why does a thick forest tend to keep the air over and around it fresher to breathe than a bare, treeless plain by day?

- (A) Its many plants take in carbon dioxide and release oxygen as they make food, keeping the air's gases in balance
- (B) Its trees suck in all the air's dust and then push out pure nitrogen instead
- (C) Its tall trees physically block stale air from ever drifting over the forest
- (D) Its animals breathe out far more oxygen than the plants ever use up

46. A hill village above an intact forest enjoys a stream that flows all year and rarely floods. After the forest below is felled, which prediction about the stream is best?

- (A) It will flow even more steadily, because the missing trees used to drink it dry
- (B) It will tend to flood hard in the rains and shrink or dry up in the dry months, because rain now runs off fast instead of soaking in
- (C) It will stay exactly the same, since trees have no effect on how water moves
- (D) It will turn into a deep lake, because the bare soil now soaks up all the rain

47. A single large forest tree falls and slowly rots over several years. What becomes of the nutrients that were locked inside its trunk and branches?

- (A) They are lost from the forest forever the moment the tree stops living
- (B) They rise up into the canopy of the neighbouring living trees
- (C) They are turned directly into sunlight that other plants then use
- (D) Decomposers gradually release them back into the soil, where new plants can take them up

48. Which arrow diagram correctly shows the direction in which ENERGY flows through a forest?

- (A) Tiger → deer → grass → Sun
- (B) Grass → Sun → deer → tiger
- (C) Deer → grass → tiger → Sun
- (D) Sun → grass → deer → tiger

49. Aman says a forest is worth keeping only for the timber it can give. Bina says it also gives oxygen, holds the soil, recharges water and shelters wildlife. Whose view is correct and why?

- (A) Bina, because besides products like timber a forest provides many life-supporting services at the same time
- (B) Aman, because timber is the only thing a forest can really provide to people
- (C) Both are wrong, because forests give neither products nor any useful services
- (D) Neither view counts, because a forest's only value is as a home for tigers

50. A forest officer must choose the action that best CONSERVES a forest and the services it provides. Which is the best choice?

- (A) Clear the forest for a single crop so the land earns money each season
- (B) Remove the decomposers so that nothing rots and the forest stays clean
- (C) Protect the standing forest and plant trees on land where forest has been lost
- (D) Cut down the canopy trees so more sunlight can reach the forest floor

51. A student writes: "During the day a forest releases oxygen, so a forest can never take in any oxygen at all." What is the precise correction?

- (A) It is fully correct — plants only ever release oxygen and never take any in
- (B) Plants take in oxygen only and never release any, so the student has it backwards
- (C) Forest plants do also take in oxygen for their own respiration, but by day they release much more oxygen than they use, so the air gains oxygen overall
- (D) Plants neither take in nor release any gas; they only soak up sunlight

52. A naturalist says, "The forest floor has the most living things crawling and growing in it, yet it is also the layer with the least light." How can both be true?

- (A) It cannot be true; the layer with the least light must always have the fewest living things
- (B) Most light is captured by the layers above, but the floor's piled-up dead matter and damp shelter support huge numbers of decomposers and ground organisms
- (C) The floor must actually be the brightest layer, since it holds the most life
- (D) The animals there glow with their own light, which is why they need none from the Sun

53. Forest web: clover → caterpillar → frog → snake, and clover → mouse → snake. The snake therefore eats both the frog and the mouse. If a drought kills almost all the MICE, what is the most likely first effect on the snake?

- (A) The snake dies out at once, because the mouse was its only possible food
- (B) The snake loses one of its two foods and must rely more on frogs, so its food supply is squeezed but not cut off completely
- (C) The snake is completely unaffected, because it never really ate the mice
- (D) The snake's numbers rise, because fewer mice leaves more clover for the snake

54. In the same web (clover → caterpillar → frog → snake; clover → mouse → snake), the snake eats the frog and the mouse, the frog eats the caterpillar, and the caterpillar and mouse eat the clover. Which organism is the producer that everything here ultimately depends on?

- (A) The snake, because as the top animal it produces food for those below it
- (B) The frog, because it sits in the middle and feeds both the levels around it
- (C) The caterpillar, because it is the first animal and so it is the producer
- (D) The clover, because it is the green plant that makes the food the whole web runs on

55. Why does protecting a large, varied forest also protect a great VARIETY of living things (its biodiversity)?

- (A) Because a forest contains only one kind of organism, which is easy to protect
- (B) Because the variety of life comes from the soil minerals, not from the forest itself
- (C) Because animals can survive equally well on bare land, so the forest is not really needed
- (D) Its many layers and habitats offer homes and food to many different species, so keeping the forest keeps the species that depend on it

56. A grassland with scattered bushes is gradually colonised over many decades — first more herbs, then shrubs, then young trees that close into a canopy, and finally a layered community with rich humus and varied animals. This slow change shows that a forest is:

- (A) Fixed and unchanging once the first bushes appear on the land
- (B) Dynamic — it develops and renews through stages over a long time rather than appearing fully formed

- (C) Entirely man-made, since only people can plant the trees in the right order
- (D) Able to reach its full, layered form within a single growing season

57. A teacher draws soil → roots take up nutrients → plant tissue → eaten by animals → dung and dead bodies → broken down → back to soil, and calls the whole loop a 'cycle' rather than a 'chain'. Why is 'cycle' the better word here?

- (A) Because the nutrients move in a perfect circle of exactly the same size every year
- (B) Because the same nutrients are used again and again, returning to the soil to be taken up once more, instead of ending at one point
- (C) Because a chain can only ever have three organisms while a cycle can have more
- (D) Because the nutrients are destroyed at the end and a fresh set is made each time

58. Forest floor microorganisms, fungi and bacteria are sometimes called the forest's 'recyclers'. The single best reason for this title is that they:

- (A) Break down dead plants and animal dung into humus, returning the locked-up nutrients to the soil for reuse
- (B) Make fresh food from sunlight on the floor, supplying the producers of the forest
- (C) Carry living leaves and animals down into the soil to store them safely
- (D) Remove nutrients permanently from the forest so the soil stays clean and bare

59. Web: grass → zebra → lion, and grass → zebra → hyena, with hyenas and lions both also feeding on each other's leftover kills. If the GRASS is destroyed by a long drought, what is the most far-reaching effect, and why?

- (A) Only the zebra is affected, because the lion and hyena eat meat, not grass
- (B) Only the lion is affected, because it sits alone at the very top of the web
- (C) Every animal in the web is affected, because all of them depend, directly or through the zebra, on the grass as the producer
- (D) Nothing is affected, because the lions and hyenas can make their own food

60. Putting the whole chapter together, which single statement best explains why a forest is called "our lifeline"?

- (A) It only gives us timber and firewood that we can cut and sell
- (B) It supplies oxygen, recycles nutrients, holds soil and water, helps balance the climate and shelters countless species — all at once
- (C) It matters only to the wild animals living in it and not to people at all
- (D) It is just a large group of trees standing together with no wider use

Solutions — Science (Class 7), Chapter 17: Forest: Our Lifeline — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	C	21	D	31	D	41	C	51	C
2	D	12	C	22	A	32	D	42	A	52	B
3	D	13	D	23	B	33	A	43	B	53	B
4	A	14	B	24	B	34	C	44	B	54	D
5	C	15	A	25	A	35	A	45	A	55	D
6	A	16	D	26	C	36	D	46	B	56	B
7	B	17	A	27	D	37	C	47	D	57	B
8	B	18	A	28	B	38	D	48	D	58	A
9	A	19	D	29	C	39	A	49	A	59	C
10	B	20	C	30	C	40	C	50	C	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) What makes something a forest is being a stable, self-renewing community of many different kinds of organisms arranged in layers, lasting for years — not any one ingredient. A crowd of a single tree species is more like a plantation; one predator alone does not make a forest; and grass alone is just a grassland. Only the multi-species, multi-layered, long-lasting community captures the idea of a forest as a living system.

2. (D) The combined leafy roof made by the tops of many tall trees together is the canopy. The branchy top of a single tree on its own is that tree's crown. So the whole-forest feature is the canopy and the single-tree feature is the crown — the order Riya then Sam matches canopy then crown.

3. (D) Plants growing in the shaded zone below the tallest trees but above the ground make up the understorey, and such plants are adapted to survive on the reduced light filtering down. They are not in the canopy (that is the top roof), the floor (that is the ground itself), and 'crown' names the top of an individual tree, not a forest layer.

- 4. (A)** Decomposers such as fungi and bacteria continually convert the fallen litter into humus, so leaf fall and leaf breakdown roughly balance and no permanent pile builds up. Leaves are not blown away wholesale, trees do not re-absorb shed leaves, and leaves do not vanish instantly on landing — they take time to be broken down.
- 5. (C)** Humus is the nutrient-rich material decomposers make from dead matter, and its key role is enriching the soil so plants can absorb those nutrients again. It is not a source of breathable oxygen, it does not waterproof the soil (it actually helps soil hold and soak up water), and it feeds the soil and plants rather than serving as direct animal food.
- 6. (A)** The grasshopper eats grass, so it is a plant-eating herbivore, and the energy it uses came from the grass — a green producer. It is not a producer (it cannot make its own food), not a carnivore (it eats plants, not the frog), and not a decomposer (it eats living plant matter, it does not break down the dead).
- 7. (B)** Energy is lost at every link of a food chain, so less is available to support life higher up; that is why top carnivores are few while producers are many. The pattern is set by energy supply, not by social habits, not by an inability to reproduce, and certainly not because tigers need little food — large predators in fact need a great deal.
- 8. (B)** In a real forest, organisms have several food sources and several predators, so the simple chains interconnect into a web — that interlinking is what makes the web realistic. A web still has producers, it does not have a fixed organism count, and energy still flows one way from producers upward, not backward.
- 9. (A)** The owl eats warblers (caterpillar → warbler chain) and shrews (beetle → shrew chain), so it links two separate chains. The warbler and shrew are different animals, the owl is a carnivore not a producer, and being at the top does not mean it eats plants directly — it eats other animals.
- 10. (B)** Caterpillars feed the warblers and graze the leaves, so losing them first starves the warblers and relieves grazing pressure on leaves. Owls do not die instantly — they can still eat shrews; the leaves are not protected by caterpillars (the caterpillars ate them); and beetles eat leaves, not caterpillars, so beetles are not directly cut off.
- 11. (C)** Removing the top predator releases its prey (warblers and shrews), which can multiply and then eat more caterpillars and beetles. Owls are carnivores, not producers of leaves; their removal does matter because they controlled the mid-level animals; and owls ate the birds and shrews, not the caterpillars and beetles directly, so those plant-eaters do not rise immediately — they may instead be eaten more.
- 12. (C)** Decomposers recycle dead leaves and animals into humus, releasing nutrients back to the soil for the trees; remove them and the litter accumulates while nutrients stay trapped, starving the soil. Decomposers do not steal food from living trees, trees do need

soil nutrients (not sunlight alone), and decomposers act on dead matter, not on healthy living roots.

13. (D) Decomposers are the link that converts dead matter back into humus that the soil reabsorbs, so cutting them out is exactly where the return path breaks. Extra sunlight does not break the loop, herbivores are not what makes humus (decomposers are), and normal rain helps soak nutrients into the soil rather than carrying them all away.

14. (B) The canopy buffers the lower storeys from direct sun, heat and heavy rain; strip it away and shade-adapted plants and floor animals are suddenly exposed to conditions they cannot tolerate. The canopy is not the lower plants' nutrient source (that is the soil and humus), floor animals breathe ordinary forest air, and losing the canopy tends to make the floor hotter and harsher, not freezing.

15. (A) Producers are so called because they make food from sunlight, water and carbon dioxide, supplying the energy every consumer ultimately uses. Being most numerous is a result, not the reason; giving us wood is a human use, not the ecological role; and plants make food from CO₂ and sunlight, not by taking in oxygen and dead matter.

16. (D) Only the wild dog is listed as eating two animals — deer and foxes. The fox eats only rabbits (not deer), the deer eats shrubs (a plant, and not rabbits), and the rabbit eats herbs (a plant, and certainly not foxes), so none of those three feeds on two animals.

17. (A) Foxes eat rabbits, so losing the rabbits cuts the fox's food supply most directly. Deer eat shrubs, not herbs, so they do not compete with rabbits; the wild dog eats deer and foxes, not rabbits; and shrubs are eaten by deer, not protected by rabbits — rabbits eat herbs.

18. (A) Decomposers — fungi and bacteria — break dead bodies of any forest organism back into the soil as humus. Deer are plant-eating herbivores and do not feed on carcasses, the Sun does not recycle dead matter into soil, and rabbits eat herbs rather than transporting nutrients to roots.

19. (D) By day, forest plants photosynthesise, absorbing carbon dioxide and releasing oxygen, which is why they are likened to lungs that refresh the air. They are not taking in oxygen and giving out CO₂ like animal breathing, they do not function by storing water as mist, and they actively exchange these gases rather than just filtering dust.

20. (C) Tree roots and litter normally bind the soil and slow rainwater so it soaks in; once removed, fast surface runoff strips the unprotected soil. The cause is loss of root grip and slowing, not the Sun making soil float, not any change in the weight of rain, and not trees having 'drunk the soil.'

21. (D) Forest soil with roots and litter acts like a sponge, taking in rain and releasing it slowly to keep streams steady; without that cover, rain runs straight off, flooding then

leaving nothing stored. The effect comes from infiltration and slow release, not from trees timing clouds, not from stream depth, and bare land in fact soaks up less, not more, than forested soil.

22. (A) Forests give off water vapour by transpiration and cast shade, raising local humidity and supporting rainfall, so losing them makes the area hotter and drier. Trees do not physically pull clouds down with their trunks, healthy trees do not burn to seed clouds, and they moderate rather than completely absorb the Sun's heat.

23. (B) Living trees absorb carbon dioxide as they photosynthesise; remove many of them and less CO₂ is taken up, so it builds up. Oxygen actually falls because there are fewer plants releasing it, the gas at issue is carbon dioxide not nitrogen, and the question is about the gas that rises because trees that used to absorb it are gone.

24. (B) Plants take in carbon dioxide and lock the carbon into their wood and leaves, and as litter breaks down some passes into the soil as humus — so wood, leaves and soil are the main stores. It is not held in the open air (that is the source it came from), not in the streams, and not chiefly in animal fur and feathers.

25. (A) Honey, gum, medicinal herbs and timber all come directly from forest plants and animals. The other lists mix in plastic, petrol, glass and cement — manufactured or mineral products that are not gathered from a forest — so only the all-forest list is fully correct.

26. (C) Forest regrowth is a slow, step-by-step recovery — herbs first, then shrubs, then trees, with soil and the web of species rebuilding over many years; it cannot be rushed into a few months. It is not done in one season, it is not true that it never recovers, and no single animal can make it regrow overnight.

27. (D) Without their predator, deer multiply and overgraze the seedlings and shrubs the forest needs to regenerate, so the whole system suffers — showing how removing one species disturbs many. Extra dung does not instantly create forest, plants and animals are tightly interdependent (not unaffected), and herbivorous deer do not hunt tigers.

28. (B) A forest is a web of interdependent organisms, so altering one species sends effects along its many feeding and sheltering links to others. The opposite ideas — that species are independent, identical, or that only the biggest one matters — all deny the interconnection that actually causes the ripple.

29. (C) What separates a forest from a wheat field is that the forest is a long-lasting community of many different species arranged in layers, while the field is one crop species grown briefly and removed. The difference is not mainly size, fields do contain some animals, and the test is the lasting multi-species community, not the presence of a single tree.

30. (C) The snake eats the small bird, so it is a flesh-eating carnivore at the chain's top, and the energy in it was originally captured by the leaves and passed up the chain. It does not eat the leaves itself (so not a herbivore), it cannot make its own food (so not a producer), and it eats living prey rather than breaking down the dead (so not a decomposer).

31. (D) A mushroom is a fungus, and fungi are decomposers, so that pairing is right. A tiger is a meat-eating carnivore, not a producer; grass is a green producer, not a carnivore; and a fungus decomposes dead matter rather than grazing on plants like a herbivore.

32. (D) The many services flow from the forest being one interconnected living system in which plants, animals, soil and microbes each support the rest. It is not just stored timber, not a set of unrelated trees, and not a man-made structure needing yearly rebuilding — those views miss the working web that produces the services.

33. (A) Forest soil is enriched by decomposers continually making humus from litter, and held in place by roots and leaf cover, whereas a bare hillside loses both the humus supply and the soil itself to runoff. Fertility is not caused by blocking all rain, by hoof-ploughing, or by mere darkness.

34. (C) Removing the forest removes the very things that hold soil, support rainfall, house wildlife and absorb CO₂, so erosion, drier and less reliable rain, habitat loss and rising CO₂ all follow. The other options reverse reality — deforestation lowers oxygen, makes streams flashier, dries the climate and worsens flooding rather than improving any of these.

35. (A) Tree-top dwellers like monkeys and fruit-eating birds live in the canopy, the high, sunlit layer rich in leaves and fruit. The forest floor is the shadiest layer with the least light, the understory is the middle layer not the treetops, and these animals live among branches, not buried in the soil.

36. (D) The floor accumulates litter and dead matter and stays damp and shaded, exactly what worms, beetles and fungi need to feed and live. It is the darkest, not the sunniest layer, oxygen is present throughout the forest not just on the floor, and the floor is not freezing — decomposers thrive in mild, moist conditions.

37. (C) The filtered light lets shade-adapted lower plants carry out enough photosynthesis to survive, so it is essential rather than harmful. It is not something to block out, it does feed the plants and not just warm animals, and light does not remove a plant's need for water and soil nutrients — it works together with them.

38. (D) Light is intercepted layer by layer from the top down, so the forest floor gets the least and is where shade plants and decomposers concentrate. The canopy is the topmost, best-lit layer (not shaded by clouds), the layers do not get equal light, and the understory is the middle layer, not buried in soil.

39. (A) A mould fungus is a decomposer; its job is breaking dead matter into humus that enriches the soil. A deer is a living-plant-eating herbivore, a mango tree is a producer (it makes food, it does not decompose anything), and a hawk is a carnivore that hunts live prey — none of these decomposes dead matter.

40. (C) Both webs start at the grass, a green producer that traps sunlight, so every animal in either web ultimately draws its energy from that producer. Energy does not start at the top predators and flow down, animals cannot make their own energy, and animals get energy by eating, not by feeding directly on humus.

41. (C) Energy shrinks at every link, so the organism farthest from the producer — the hawk at the top — relies on the smallest remaining share. It is not about body size (the grasshopper is near the bottom and gets a larger share), the grass actually captures the most energy as the producer, and the frog in the middle still gets more than the hawk above it.

42. (A) The forest keeps its soil fertile because decomposers continuously recycle nutrients from dead matter back into the soil, replacing what growing plants withdraw. Plants do not create nutrients from sunlight, the nutrients are genuinely taken up and built into tissue (not merely borrowed), and rain does not deliver a daily fresh nutrient supply from the sky.

43. (B) Because one decomposer community recycles the dead from both the pond and the land chains, those chains are linked through the shared cycling of nutrients into a common soil. This shows water and land chains are connected, not separate; decomposers act on both aquatic and terrestrial dead matter; and they are not partitioned chain-by-chain.

44. (B) The fish feed the kingfisher and graze the algae, so losing them starves the kingfisher and lets algae accumulate. The land deer and tiger do not eat fish, so they are not directly hit; the algae are eaten by the fish (the fish are not the algae's food); and the kingfisher cannot increase when its food disappears.

45. (A) By day the forest's plants photosynthesise, drawing in carbon dioxide and giving off oxygen, which keeps the air balanced and fresh. They do not work by turning dust into nitrogen, tall trees do not wall off stale air, and animals breathe out carbon dioxide (not oxygen) — it is the plants that supply the oxygen.

46. (B) Without the forest's sponge of roots and litter, rain runs straight off — giving floods in the wet season and a dry stream later, the opposite of the steady year-round flow before. The trees did not 'drink the stream dry' (they actually steadied it), forests clearly do affect water flow, and bare soil soaks up less, so no lake forms.

47. (D) As decomposers break the fallen trunk down over the years, the nutrients in the wood pass back into the soil for new plants to reuse. They are not lost forever (the cycle

reclaims them), they do not travel up into other trees' canopies, and matter cannot be converted into sunlight.

48. (D) Energy starts at the Sun, is captured by green grass (the producer), passes to the plant-eating deer, then to the meat-eating tiger — so Sun → grass → deer → tiger is right. The reversed arrows are wrong because energy does not flow back toward the Sun, the Sun is the source and cannot sit after the grass, and a deer cannot precede the grass it eats.

49. (A) Bina is right: a forest yields products such as timber but, just as importantly, supplies oxygen, soil protection, water recharge and wildlife habitat all together. Aman wrongly reduces it to timber, it is false that forests give nothing useful, and a forest's value extends far beyond being a tiger's home.

50. (C) Conservation means safeguarding the existing forest and restoring lost forest by planting trees. Clearing it for a crop destroys the very services in question, removing decomposers would wreck nutrient recycling and soil fertility, and felling the canopy exposes and harms the lower layers — none of these conserves the forest.

51. (C) Living plants respire and so do take in some oxygen, but during daytime photosynthesis they release far more than they consume, giving a net gain of oxygen to the air. So it is wrong to say they never take any in; equally wrong to say they only take in and never release; and false that they exchange no gases at all.

52. (B) The upper layers intercept the light, so the floor is dim, yet its accumulation of dead matter and moisture feeds vast numbers of decomposers and small ground dwellers — so abundant life and low light go together there. Abundant life does not require the most light, the floor is not secretly the brightest layer, and forest floor organisms do not make their own light.

53. (B) Since the snake eats both mice and frogs, losing the mice tightens its food supply but it can still eat frogs, so it is pressed but not wiped out. The mouse was not its only food (so it does not die at once), the snake did eat mice (so it is affected), and the snake does not eat clover, so more clover would not help it increase.

54. (D) Clover is the only green plant here, so it is the producer that makes the food every other organism in the web depends on. The snake is a top carnivore (it consumes, it does not produce), the frog is a carnivore in the middle, and the caterpillar is a plant-eating herbivore — being the first animal does not make it a producer.

55. (D) A forest's range of layers and habitats houses many species, so safeguarding the forest safeguards the diversity of life that lives there. A forest is not a single-species place, biodiversity arises from the living community rather than from minerals alone, and most forest species cannot simply move to bare land — they depend on the forest.

56. (B) The step-by-step build-up from herbs to shrubs to a layered, humus-rich forest over decades shows the forest is a dynamic system that develops and renews over long time. It is not fixed once bushes appear, it can establish naturally rather than only by human planting, and it certainly cannot reach its full layered form in one season.

57. (B) It is a cycle because the nutrients keep being reused — returned to the soil by decomposers and taken up again — rather than running out at a final link like a one-way chain. It is not about the loop being a literal equal-sized circle, not about how many organisms a chain may hold, and the nutrients are recycled, not destroyed and remade.

58. (A) They earn the title 'recyclers' by converting dead matter and dung into humus and thereby returning nutrients to the soil for plants to use again. They do not photosynthesise food from sunlight (that is the producers' job), they act on dead matter rather than carrying living things underground, and they return nutrients rather than removing them for good.

59. (C) Grass is the producer feeding the zebra, which feeds the lion and hyena, so destroying the grass starves the zebra and then the carnivores too — the whole web is hit. It is not only the zebra (the meat-eaters depend on the zebra, which depends on grass), not only the lion, and the carnivores cannot make their own food, so they are not spared.

60. (B) A forest earns the name 'lifeline' because it does many life-supporting jobs together — giving oxygen, recycling nutrients, holding soil and water, steadying climate and sheltering species. Reducing it to timber, to a place that matters only to its animals, or to a mere stand of trees all ignore the full set of services that make it a lifeline.